

RESTORING THE UNSEEN

Mathematical Models of Image Degradation and Restoration in Medical Diagnostics



DEGRADED IMAGE WITH SPECKLE NOISE

RESTORED IMAGE: MATHEMATICAL ALGORITHM APPLIED

The Anatomy of a Flawed Diagnostic Image

[Blue] Target Image (f, F)

[Orange] Blur/Degradation (h, H)

[Magenta] Chaos/Noise (η , N)

[White] Degraded Result (g, G)

Spatial Domain

$$g(x,y) = h(x,y) * f(x,y) + \eta(x,y)$$

Convolution: The mathematical 'smearing' of the image pixels.

Frequency Domain

$$G(u,v) = H(u,v) \cdot F(u,v) + N(u,v)$$

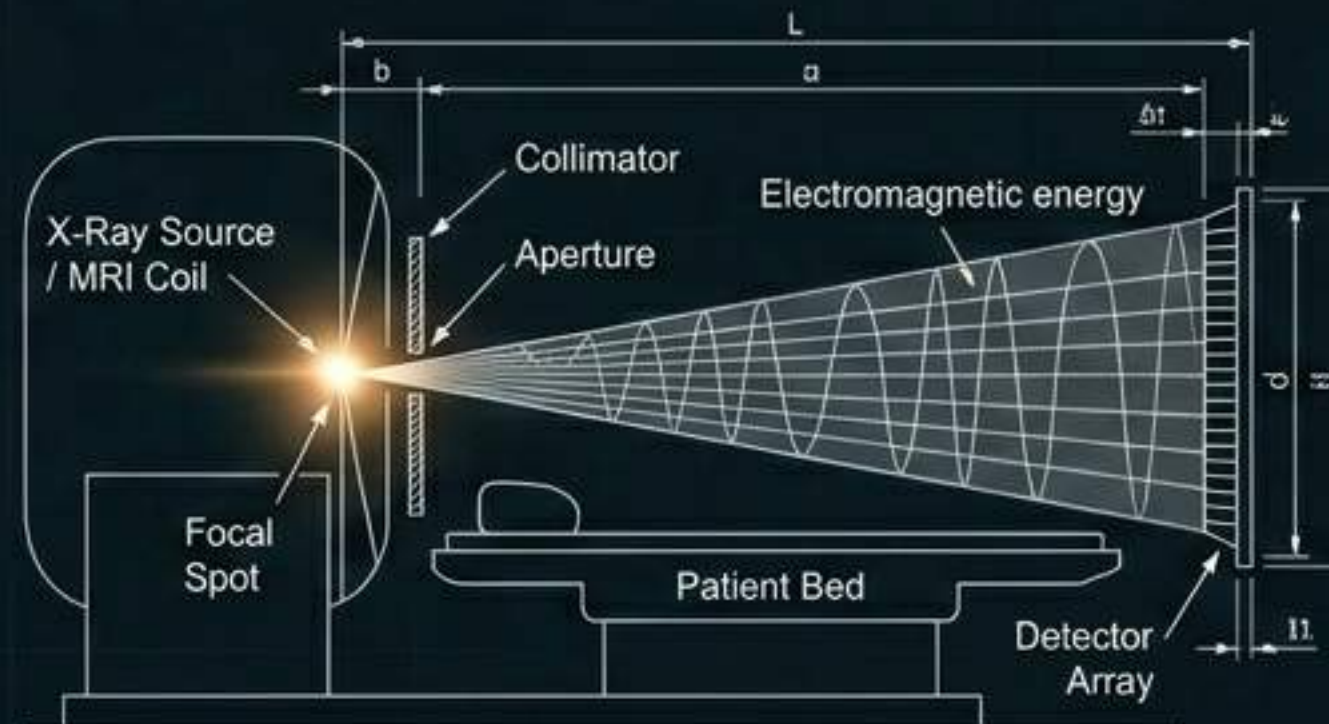
Multiplication: Convolution simplifies into direct multiplication in the frequency domain.

Blur Type I: Hardware Limitations and the Gaussian Point Spread Function

$$h(x,y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x^2 + y^2)}{2\sigma^2}\right)$$

Relative Coordinates:
Distance from the center of the kernel.

The Blur Width: Higher standard deviation equals a wider, more severe blur. Normalized in a discrete domain to sum to 1.



Clinical Context: Perfectly models the blurring caused by the limited physical dimensions of the focal spot in CT and MRI hardware.

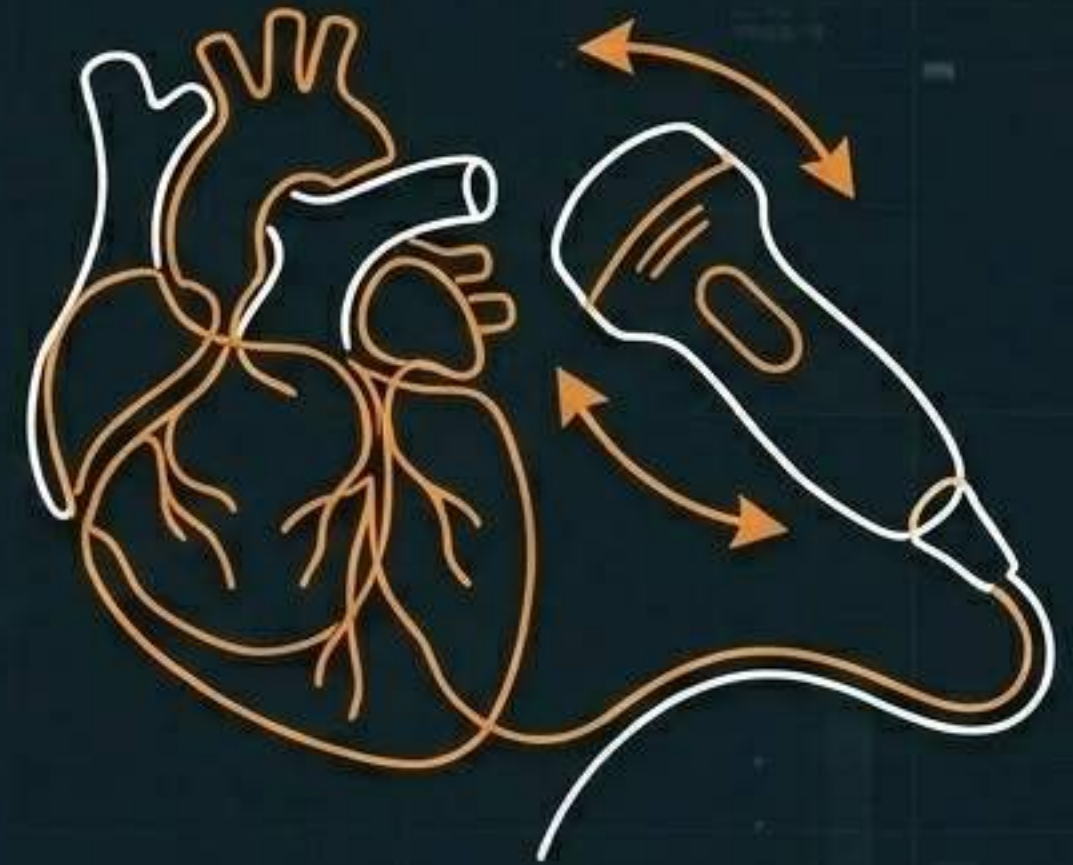
Blur Type II: Patient Movement and the Linear Motion Model

$$h(x,y) = \frac{1}{L}$$

(for $0 \leq x \leq L-1, y=0$)

Motion Length: The distance of physical movement calculated in pixels.

Motion Angle: The trajectory of the movement relative to the x-axis.



Clinical Context: Essential for correcting images distorted by involuntary patient respiration or the manual sweeping of an ultrasound probe.

The Chaos Element: Modeling Additive Gaussian Noise

$$p(\eta) = \frac{1}{\sigma_{\eta} \sqrt{2\pi}} \exp\left(-\frac{(\eta - \mu_{\eta})^2}{2 \sigma_{\eta}^2}\right)$$

Mean Noise: The baseline interference, typically assumed to be zero.

Noise Intensity: The standard deviation dictating the severity of the static or speckle.

Pristine **MRI Scan**



+ Simulated
Gaussian **Noise**



Robust Algorithm
Testing

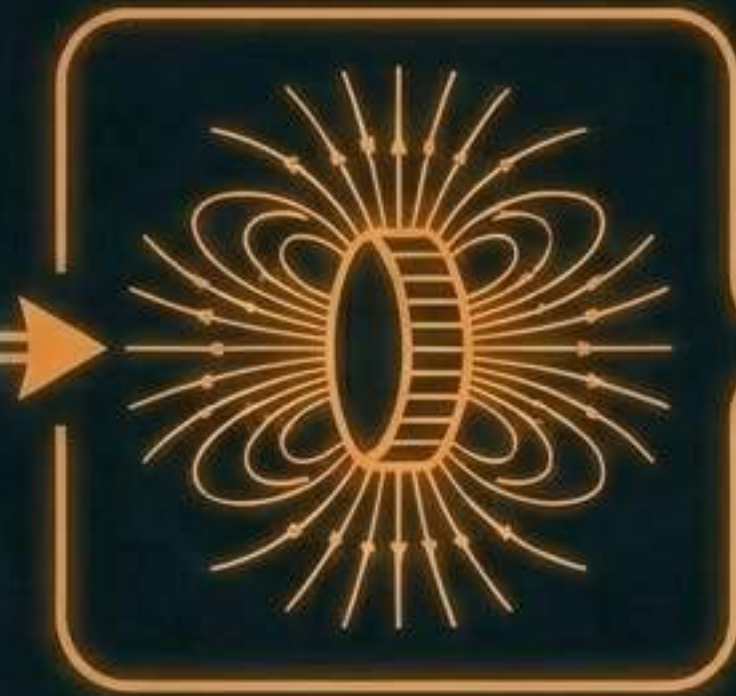
Clinical Context: Used to simulate real-world hardware conditions, allowing engineers to stress-test noise-reduction algorithms prior to clinical deployment.

The Journey of a Corrupted Diagnostic Image



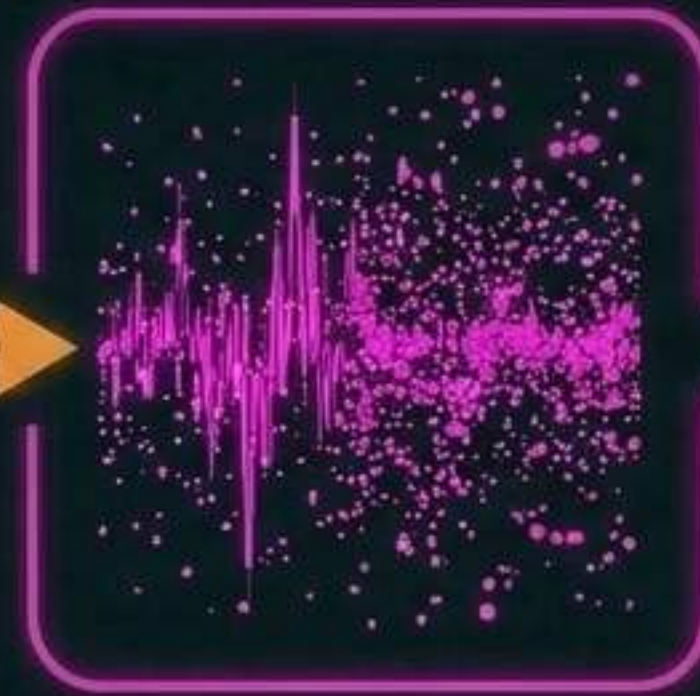
The Ideal

$f(x,y)$



The Smear

Convolution
with $h(x,y)$



The Static

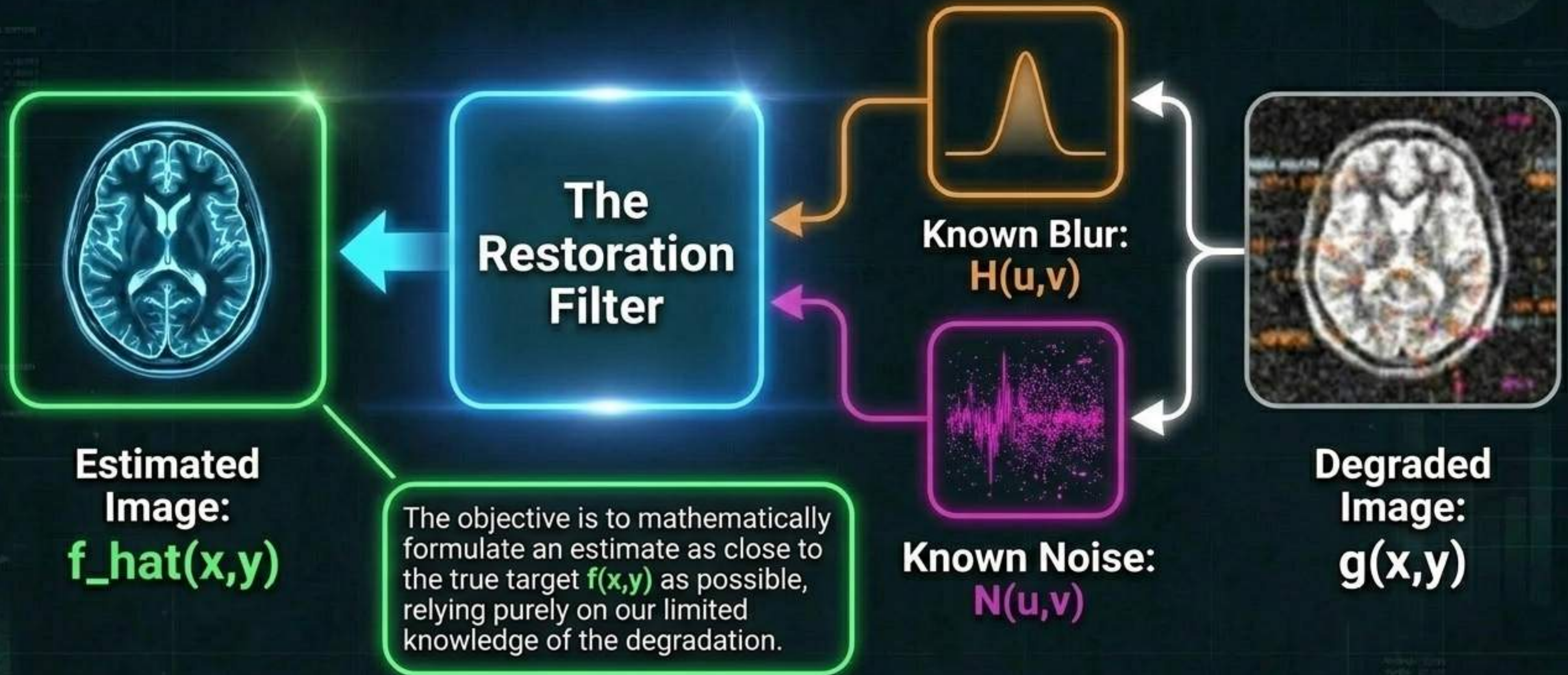
Addition of
 $\eta(x,y)$



The Output

$g(x,y)$

The Goal of Restoration: Engineering an Estimate



The Mathematical Ideal: Inverse Filtering

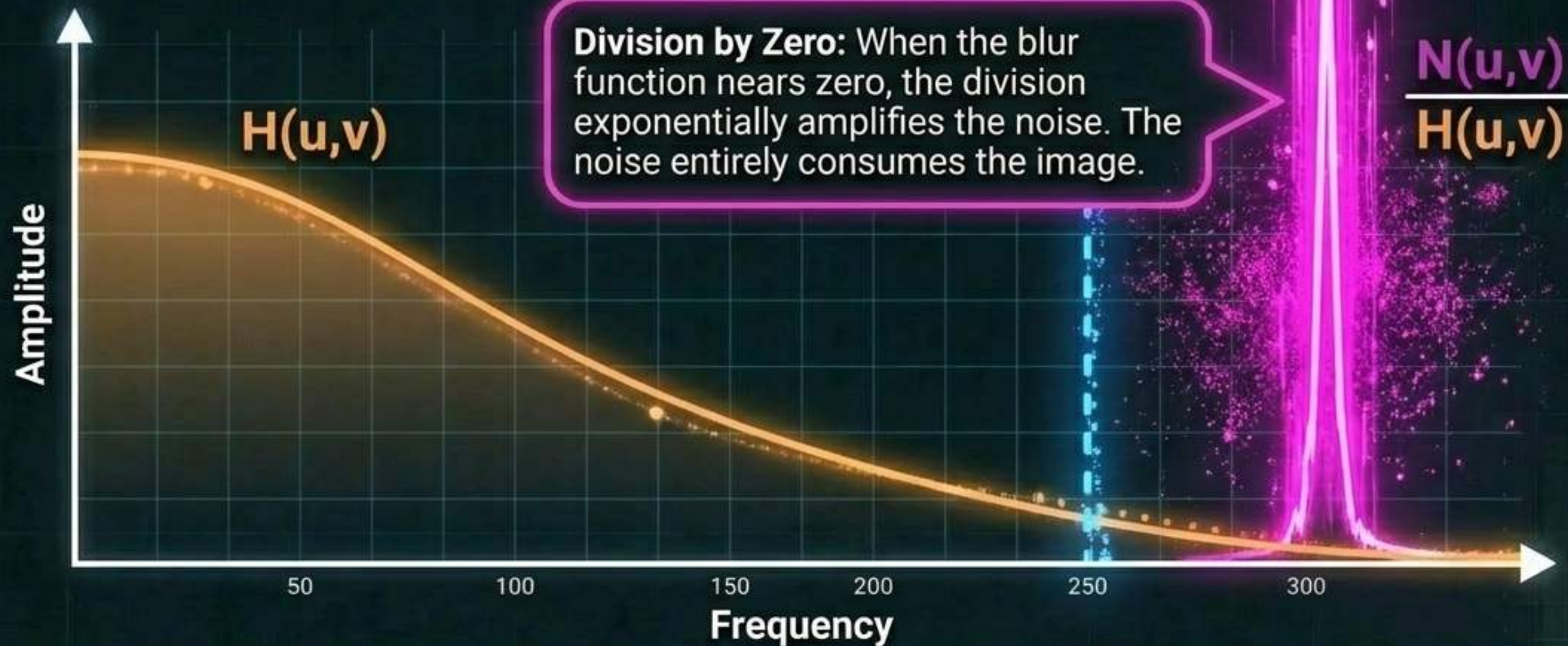
$$\hat{F}(u,v) = \frac{G(u,v)}{H(u,v)}$$

$$\hat{F}(u,v) = \frac{H(u,v) \cdot F(u,v) + N(u,v)}{H(u,v)}$$

$$\hat{F}(u,v) = F(u,v) + \frac{N(u,v)}{H(u,v)}$$

The logical first step: if the image was multiplied by a blur function, simply divide by the blur function. In a frictionless, noiseless **universe**, this perfectly recovers the original image.

The Fatal Flaw of the Inverse Filter: Noise Explosion

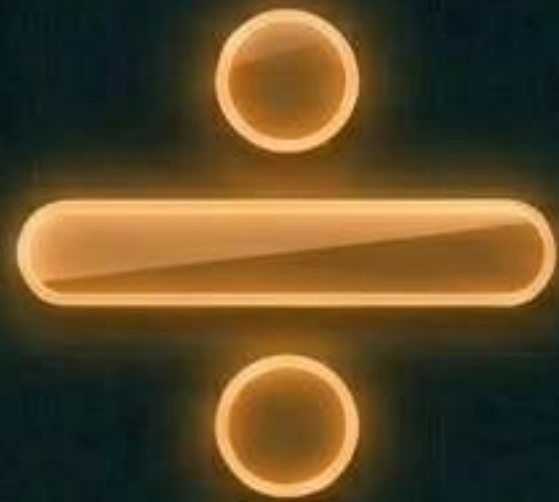


The Band-Aid Solution: Frequency Thresholding

$$Q(u,v) = \frac{1}{H(u,v)} \text{ only if } |H(u,v)| > \text{Threshold}$$

The Statistical Compromise: Wiener Filtering (MMSE)

Inverse Filter Approach



Solves for exact mathematical reversal. Fails in chaos.

Wiener Filter Approach



Solves for Minimum Mean Square Error (MMSE). Anticipates chaos.

Core Objective Equation: Minimize $e^2 = E\{(f - \hat{f})^2\}$

The Wiener filter abandons simple division. Instead, it simultaneously evaluates the blur function and the statistical properties of the noise to calculate the closest possible statistical **estimate** of the original image.

Inside the Wiener Formula: The Statistical Safety Valve

Complex conjugate of the degradation function.

$$\hat{F}_a(u, v) = \left[\frac{H^*(u, v)}{(|H(u, v)|^2 + K)} \right] \cdot G(u, v)$$

The Safety Valve: An empirical constant approximating the ratio of Noise Power to Signal Power.

The Mechanic:

When $H(u, v)$ approaches zero, the denominator no longer collapses to zero because the constant K props it up. The filter safely reduces its gain instead of amplifying the noise.

The Edge Case:

If $K = 0$ (assuming a universe with absolutely zero noise), this formula mathematically reverts back to the naive Inverse Filter.

The Mathematical Compromise: Constrained Least Squares (CLS)

The Wiener Bottleneck



Wiener requires knowing the statistical power spectral density of the uncorrupted image. In real-world diagnostics, we rarely possess this prior knowledge.

The CLS Workaround



CLS only requires the mean and variance of the noise. These metrics are easily estimated from the blank background regions of any degraded scan.

Core Objective: Mathematical Smoothness

Instead of relying on inaccessible signal statistics, CLS seeks an image estimate that is mathematically smooth, while remaining strictly constrained by the known degradation model.

Inside the CLS Formula: Enforcing Mathematical Smoothness

The Smoothness Enforcer: The Fourier transform of the discrete Laplacian operator (a high-pass filter). It aggressively penalizes jagged, noisy artifacts.

The Lagrange Multiplier: A tuning parameter iteratively adjusted until the estimated noise perfectly matches the actual noise constraint.

$$\hat{F}_a(u, v) = \left[\frac{H^*(u, v)}{|H(u, v)|^2 + \gamma |P(u, v)|^2} \right] \cdot G(u, v)$$

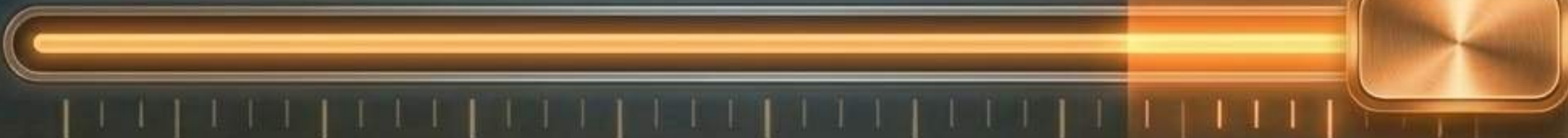
The Key Takeaway:

By swapping the statistical constant (K) for a mathematical operator (P), CLS entirely bypasses the need for the original image's power spectrum.

Wiener vs. CLS: The Prior Knowledge Trade-off

Wiener Filter

High Data Requirement

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- **Inputs needed:** Degradation Function (H), Noise Power Spectrum, Original Signal Power Spectrum.
 - **Result** Highly robust to severe noise, but often requires guessing the constant K .

CLS Filter

Low Data Requirement

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- **Inputs needed:** Degradation Function (H), Noise Mean & Variance.
 - **Result:** Requires more computational complexity to iteratively solve for gamma, but operates entirely on readily available diagnostic data.

The Restoration **Decision Matrix**

Filter Method	Formula Core	Required Data	Strategic Trade-offs
Inverse Filter	$\hat{F}_a(u, v) = \frac{G(u, v)}{H(u, v)}$	H(u,v) only	Advantage: Simple and computationally fast. Limitation: Highly unstable; fatally amplifies noise near zero.
Wiener (MMSE) Filter	Uses $\frac{H^*(u, v)}{ H(u, v) ^2 + \frac{S_{eta}(u, v)}{S_{ff}(u, v)}}$	H(u,v) + Signal & Noise power spectra	Advantage: Highly robust against severe noise. Limitation: Requires prior knowledge of pristine image statistics.
Constrained Least Squares (CLS)	Uses $\frac{H^*(u, v)}{ H(u, v) ^2 + \gamma P(u, v) ^2}$	H(u,v) + Noise mean & variance	Advantage: Relies only on easily accessible noise data. Limitation: Higher computational complexity due to iterative tuning.